Discrete Structures: Basic Structures – P2: Functions

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Outline For This Chapter and Lecture

- Basic Structures
  - Set
  - Function
    - Intro
      - Each package is assigned to one and only one truck (truck 1-4) based on its ZIP code.
        - All input ZIP code should be handled.
        - A fixed truck index should be the output.
    - Definition
    - Representation
    - Inverse Function
  - Sequence
  - Summation

Introduction 1

- In Programming in C++ class, you have learned about basics of C++
  - E.g. How to generate the grade letter (char type) based on the score (float type)

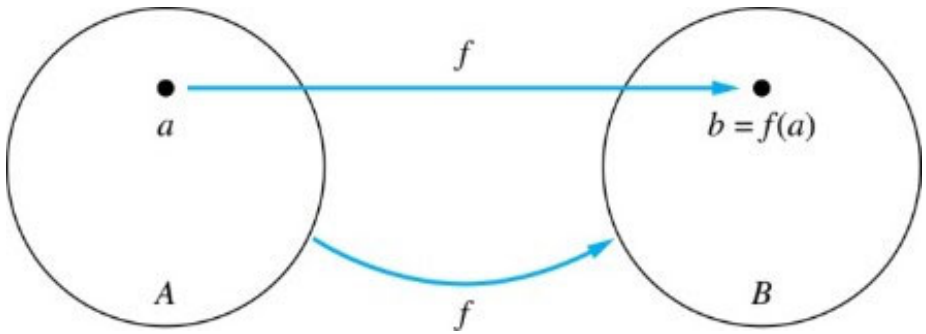
```
char generateGradeLetter(float score) {
    if (score < 60) { //covers score < 0
        return 'F';
    }
    if (score < 70) {
        return 'D';
    }
    ……
}
```

Introduction 2

- Common properties of those functions:
  - All the input cases are handled
  - There is only one output for a specific input
    - The result is **predictable**
    - Random generator can only generate “pseudo random numbers”: it’s also a function, and if we know the seed which is the input to this random generator, the result is still predictable
- Based on these common properties, how to formally define a function in math?
  - Input? Output?
  - The functionality?

Definition of Functions 1

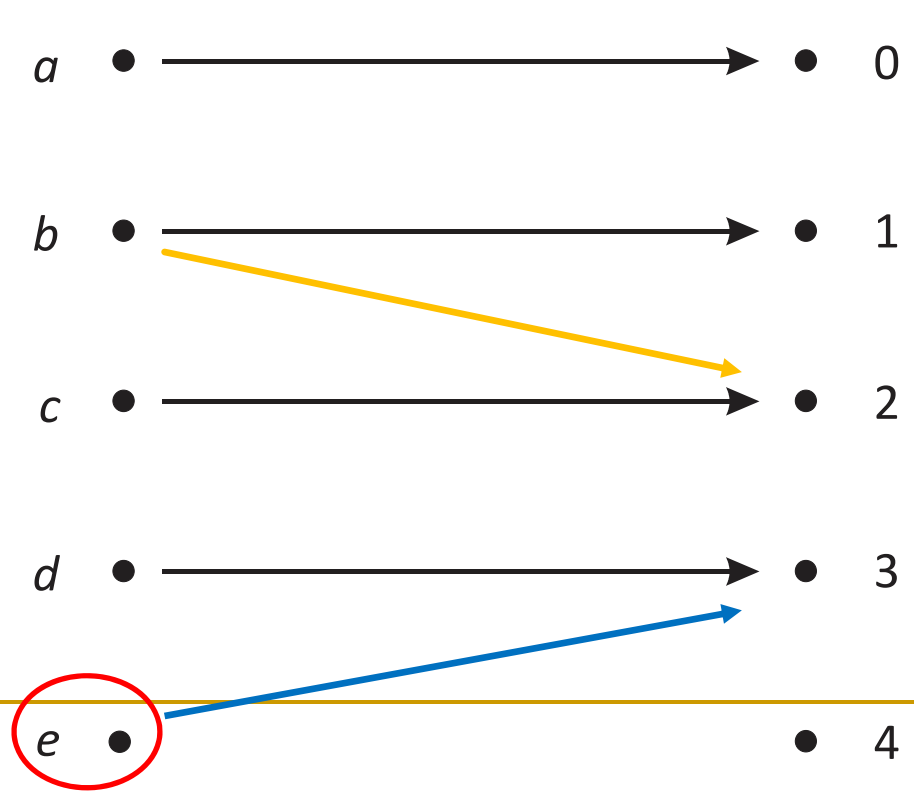
- **Definition**: Let *A* and *B* be nonempty sets. A function f from *A* to *B* is an assignment of **exactly one** element of *B* to **each** element of *A*.
  - If f is a function from *A* to *B*, we write $\boldsymbol{f: A \to B}$.
  - We write $\boldsymbol{f(a) = b}$ if *b* is the unique element of *B* assigned by the function f to the element *a* of *A*.



- In another way:
  - Full coverage on the input
  - No one-to-many mapping: predictable (only one) output

Definition of Functions 2

- **Example**: Let $A = \{a, b, c, d, e\}$, $B = \{0,1,2,3,4\}$,: $f(a) = 0, f(b) = 1, f(c) = 2, and\ f(d) = 3$. Is $f: A \to B$ a function?
  - **Solution**: the mappings can be illustrated as



1. Input e is not covered, i.e., no output assigned for input e. So the condition "full coverage of the input" is not satisfied. Then $f: A \to B$ is not a function.

2. Can be fixed by the blue arrow.

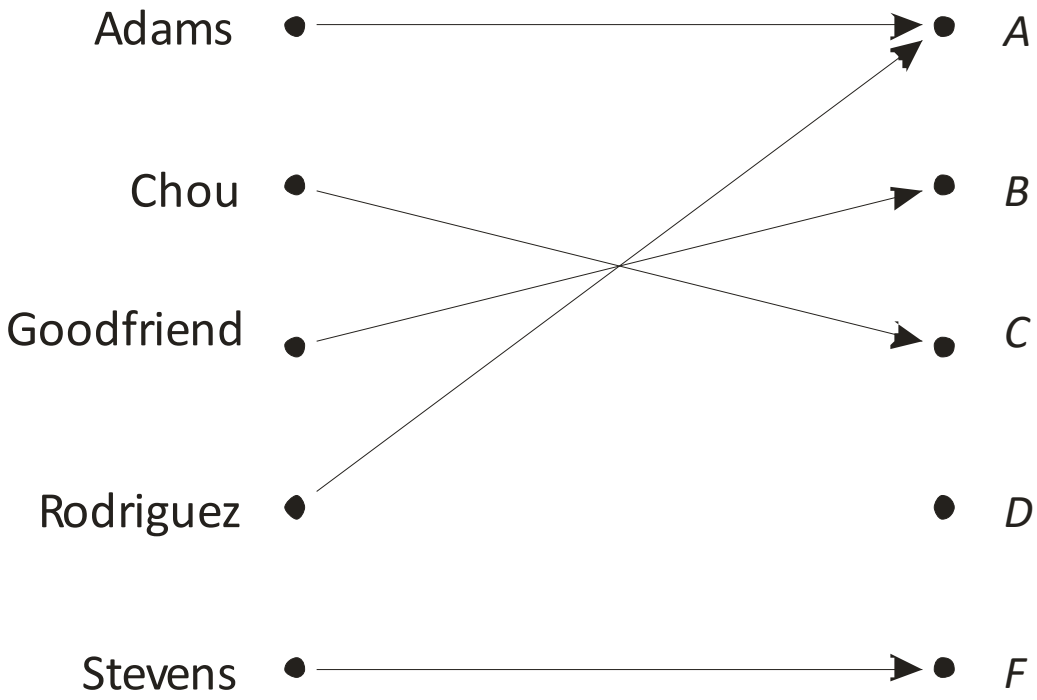
3. A one-to-many case: not a function any more.

Definition of Functions 3

- If f is a function from *A* to *B*, we say that *A* is the ***domain*** of f and *B* is the ***codomain*** of f.
- If $f(a) = b$, we say that *b* is the ***image*** of *a* and *a* is a ***preimage*** of *b*.
- The ***range*** of f is the set of all images of elements of *A*. Also, if f is a function from *A* to *B*, we say that f ***maps*** *A* to *B*.
  - By set builder notation? For a function $f: A \to B$, the range of f can be specified as $\{b | a \in A \wedge f(a) = b\}$
- The difference between codomain and range?
  - **Range ⊆ codomain**

Definition of Functions 4

- **Example**: What are the domain, codomain, and range of the function that assigns grades to students shown in the following graph?



Domain = {Adams, Chou, Goodfriend, Rodrguez, Stevens}

Codomain = {A,B,C,**D**,F}

Range = {A,B,C,F}

Representing Functions 1

- Functions can be specified in different ways:
  - An explicit statement of the assignments
    - E.g. specify the age of each student in previous example
    - $f(Abdul) = 22$, $f(Brenda) = 24$, $f(Carla) = 21$, $f(Desire) = 22$, $f(Eddie) = 24$, $f(Felicia) = 22$
  - A relation from *A* to *B*
    - E.g. $f: A \to B$ where $A = \{a, b, c, d\}$, $B = \{0,1,2,3,4\}$ can be specified as the relation $R = \{(a, 0), (b, 1), (c, 2), (d, 3)\}$
    - A pair (u, v) in this relation means $f(u) = v$
  - A formula, e.g. $f(n) = n + 1$ for $n \in \mathbb{Z}$
    - This formula can be illustrated by its graph: a discrete version
  - A real computer program / function / method
    - E.g. `int getMax(int[] nums, int size) {……}`

Representing Functions 2

- **Example**: Let *R* be the relation consisting of ordered pairs (Abdul, 22), (Brenda, 24), (Carla, 21), (Desire, 22), (Eddie, 24), and (Felicia, 22), where each pair consists of a graduate student and the age of this student. Specify a function determined by this relation?
  - **Solution**: $f(Abdul) = 22$, $f(Brenda) = 24$, $f(Carla) = 21$, $f(Desire) = 22$, $f(Eddie) = 24$, $f(Felicia) = 22$
    - Domain = {Abdul, Brenda, Carla, Desire, Eddie, Felicia}
    - Codomain = $\mathbb{Z}^+$
    - Range = {21, 22, 24}

More Topics on Functions

- We will talk about these topics on functions
  - Compound functions
  - One-to-one
  - Onto
  - Inverse functions
  - Graph of a function

Compound Functions 1

- Let f_1 and f_2 be functions from A to $\mathbb{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbb{R}$ defined by
  - $\boldsymbol{(f_1 + f_2)(x) = f_1(x) + f_2(x)}$
  - $\boldsymbol{(f_1 f_2)(x) = f_1(x) f_2(x)}$
- **Example**: Let f_1 and f_2 be functions from $\mathbb{R}$ to $\mathbb{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$
  - **Solution**:
    $(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$
    $(f_1 f_2)(x) = f_1(x) f_2(x) = x^2(x - x^2) = x^3 - x^4$

Compound Functions 2

- Let g be a function from the set *A* **to the set *B*** and let f be a function **from the set *B*** to the set *C*. The ***composition*** of the functions f and g, denoted for all $a \in A$ by $f \circ g$, is defined by
  $(\boldsymbol{f} \circ \boldsymbol{g})(\boldsymbol{a}) = \boldsymbol{f}(\boldsymbol{g}(\boldsymbol{a}))$
  - **The output of g is the input of f**

$a \rightarrow g() \rightarrow f() \rightarrow ?$

A pipeline of functions

- **Example**: Let *f* and *g* be the functions from $\mathbb{Z}$ to $\mathbb{Z}$ defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g? What is the composition of g and f?
  - $(f \circ g)(x) = f\big(g(x)\big) = \boldsymbol{f(3x+2)} = 2(3x+2) + 3 = \boldsymbol{6x+7}$
  - $(g \circ f)(x) = g\big(f(x)\big) = \boldsymbol{g(2x+3)} = 3(2x+3) + 2 = \boldsymbol{6x+11}$

Compound Functions 3

- **Example**: An item costs $100. Need to
  - f(x): add 8% sales tax, and apply
  - g(x): add 15% tips
  - $(g \circ f)(x) = g\big(f(x)\big)$ =1.15(1.08x)=124.2
  - $(f + g)(x) = f(x) + g(x)$ =100(1+0.08+0.15)=123

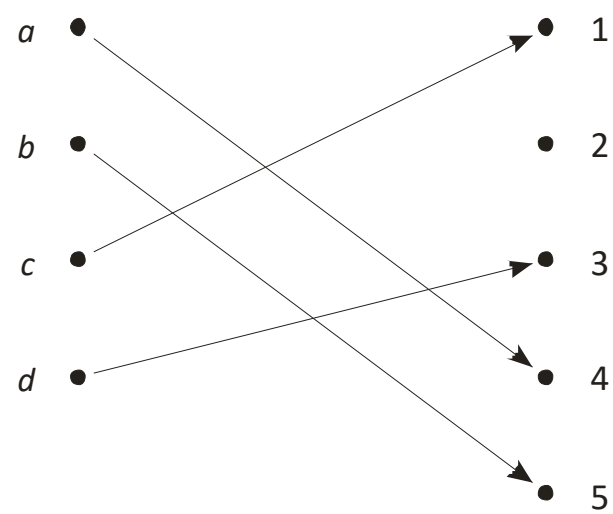
- **Example**: Let A={a, b, c, d},B={1, 2, 3}. Define f(x): A→B where f(a)=2, f(b)=3, f(c)=2, f(d)=1 g(x): B→A where g(1)=b, g(2)=d, g(3)=a.
- $(g \circ f)(x) = g\big(f(x)\big)$: A → A with {(a, d), (b, a), (c, d), (d, b)}
- $(f \circ g)(x) = f\big(g(x)\big)$: B → B with {(1, 3), (2, 1), (3, 2)}

One-to-one 1

- A function f is said to be ***one-to-one***, or an ***injunction***, if and only if $\boldsymbol{f(a) = f(b)}$ **implies that $\boldsymbol{a = b}$ for all $\boldsymbol{a}$ and $\boldsymbol{b}$ in the domain** of f. A function is said to be ***injective*** if it is one-to-one.
  - How to understand $(f(a) = f(b)) \rightarrow (a = b)$?
  - How about contrapositive? $(a \neq b) \rightarrow (f(a) \neq f(b))$
  - This means no many-to-one mapping
- In another way:
  - Full coverage on the domain / input
  - No one-to-many mapping + no many-to-one mapping -> only one-to-one mapping

One-to-one 2

- **Example**: Determine whether the mapping f from $\{a, b, c, d\}$ to $\{1,2,3,4,5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ is a one-to-one function.
  - **Solution**: First of all, this is a function covers all the input, and for each input value there is only one output value. Then there is no such a case that different input values are mapped to the same output value. So this is a one-to-one function.
  - This one-to-one function can be visualized as the right graph



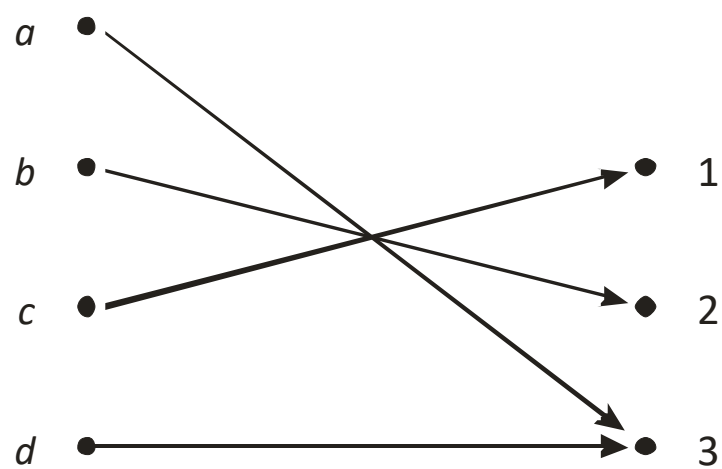
- Do you think $f(x) = x^2$ from $\mathbb{Z}$ to $\mathbb{Z}$ is one-to-one?
  - How about $f(x) = x^2$ from $\mathbb{Z}^+$ to $\mathbb{Z}^+$?

Onto 1

- A function f from *A* to *B* is called ***onto***, or ***surjection***, if and only if **for every element $\boldsymbol{b} \in \boldsymbol{B}$ there is an element $\boldsymbol{a} \in \boldsymbol{A}$ with $\boldsymbol{f(a) = b}$**. A function f is called ***surjective*** if it is onto.
  - In propositional logic: a function $f: A \to B$ is onto iff
    $$\forall b(b \in B \to \exists a(a \in A \wedge f(a) = b))$$
  - By set relations: **codomain = range**
- In another way:
  - Full coverage on the domain / input
  - No one-to-many
  - Full coverage on the codomain

Onto 2

- **Example**: Let f be the function from $\{a, b, c, d\}$ to $\{1,2,3\}$ defined by $f(a) = 3, f(b) = 2, f(c) = 1$, and $f(d) = 3$. Is f an onto function?
  - **Solution**: It's easy to verify this is a function. And the range of this function is $\{1,2,3\}$, which equals to the codomain. So this is an onto function.
  - This onto function can be visualized as the right graph



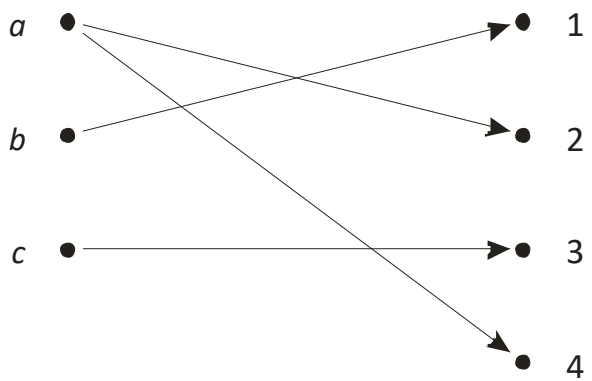
- Do you think think $f(x) = x^2$ from $\mathbb{Z}$ to $\mathbb{Z}$ is onto?
  - How about $f(x) = x^2$ from $\mathbb{Z}^+$ to $\mathbb{Z}^+$?

One-to-one and Onto 1

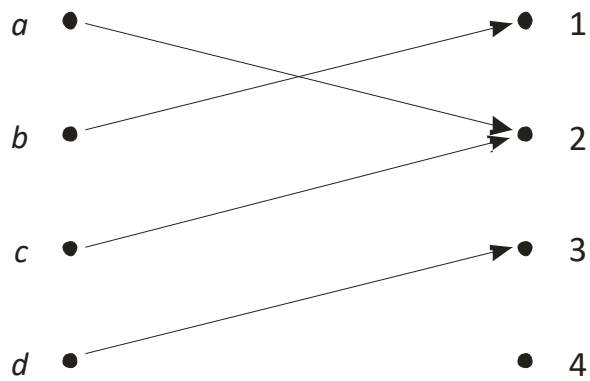
- The function *f* is a ***one-to-one correspondence***, or a ***bijection***, if it is both one-to-one and onto. A function f is called ***bijective*** if it is one.
- In another way:
  - Full coverage on the domain / input
  - No one-to-many mapping
  - No many-to-one mapping
  - Full coverage on the codomain

One-to-one and Onto 2

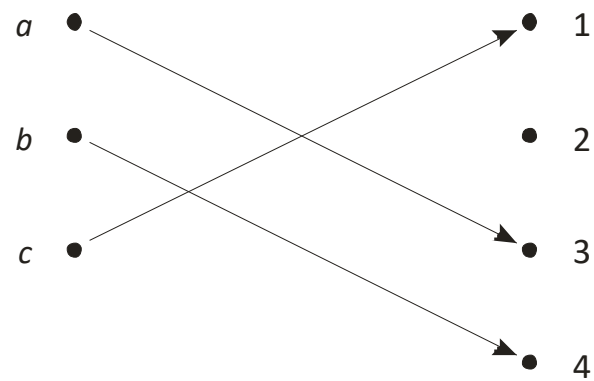
- Are the following mappings one-to-one or onto functions?



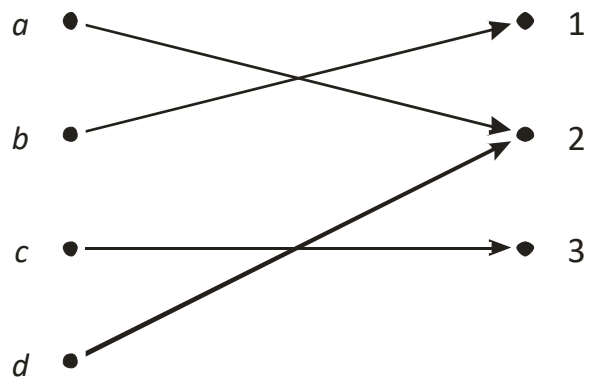
(a) Not a function



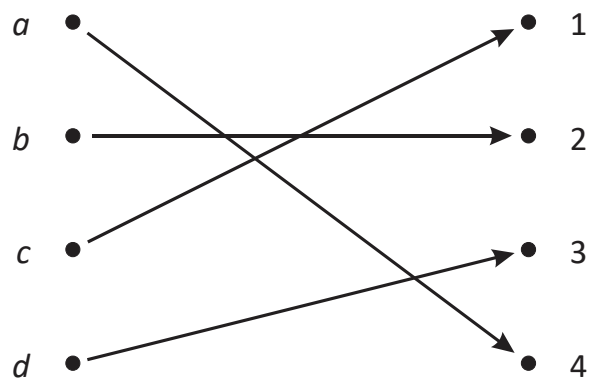
(b) Not one-to-one, not onto



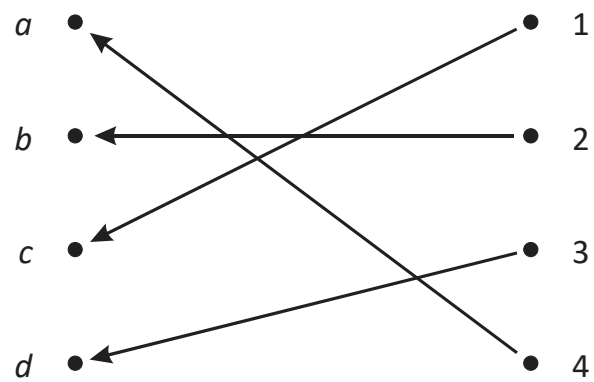
(c) One-to-one, not onto



(d) Onto, not one-to-one



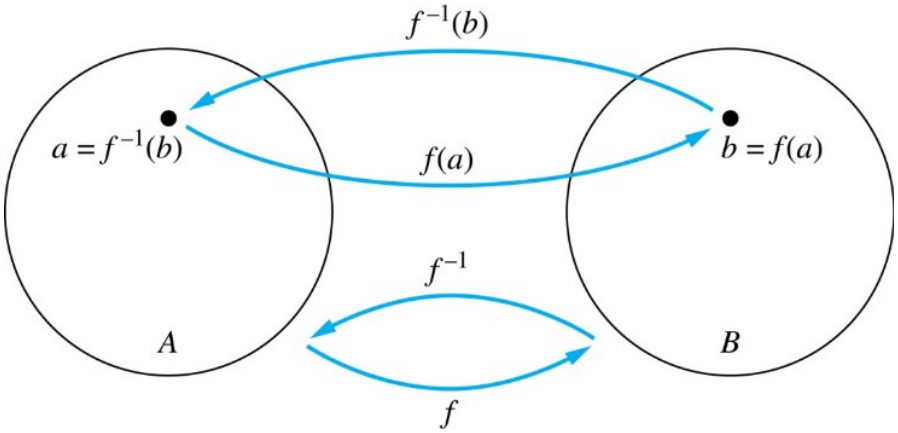
(e) One-to-one, and onto



(f) One-to-one, and onto

Inverse Function 1

- Text encryption can be specified by a function f
- We need to decrypt the code to get the text back
- The decryption procedure can be specified by another function, which is the inverse function of f, and this inverse function can be denoted as f^{-1}
- **A function f is invertible if and only if it's one-to-one correspondent**
  - Not one-to-one -> the reversed mappings could have one-to-many mappings
  - Not onto -> the reversed mappings don't have full coverage on the input

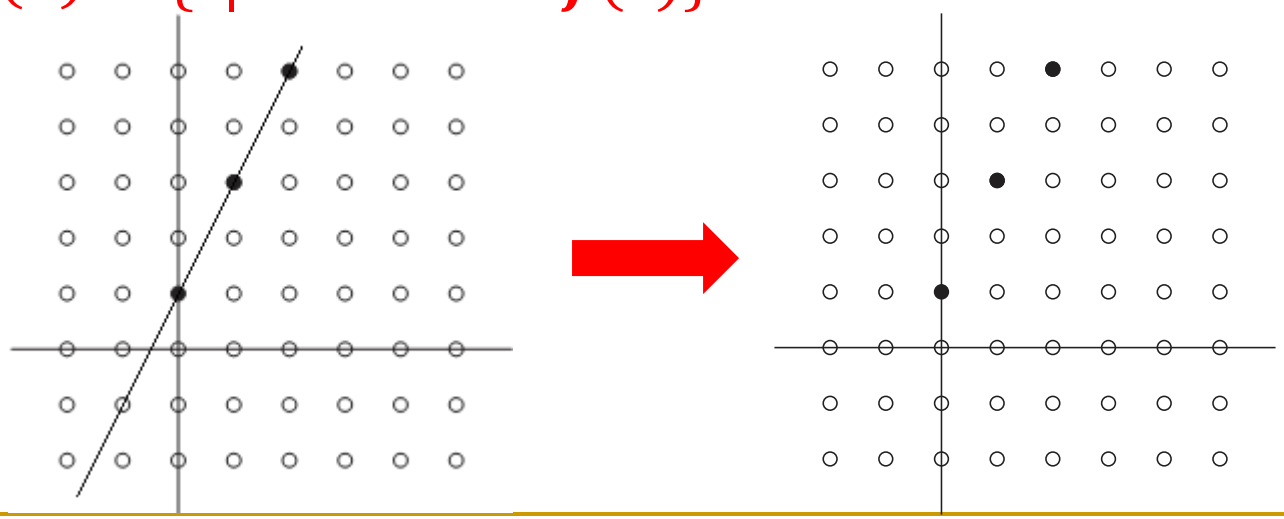


Inverse Function 2

- **Example**: Let f be the function for $\{a, b, c\}$ to $\{1,2,3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Then f is one-to-one correspondent, and f is invertible: $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$
  - Domain of f^{-1}: $\{1,2,3\}$
  - Codomain of f^{-1}: $\{a, b, c\}$
- **Example**: Let $f: \mathbb{Z} \to \mathbb{Z}$ be such that $f(x) = x + 1$. Then f is one-to-one correspondent, and thus f is invertible.
  - Suppose that *y* is the image of x, so that $y = x + 1$. Then $x = y - 1$. This is the function that can help us get x back from y, consequently, $f^{-1}(y) = y - 1$.

Graph of a Function

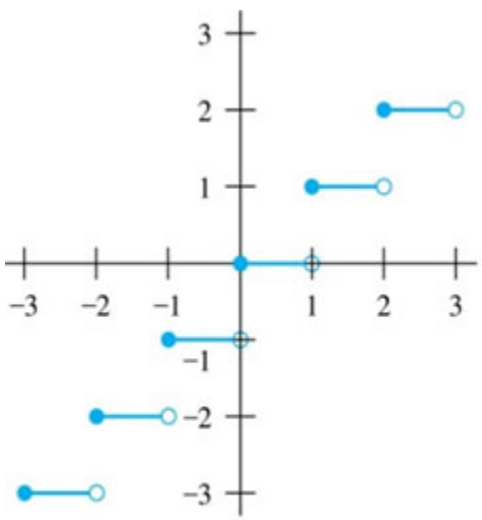
- Recall the graph a function in Calculus
  - Think about the graph of $f(x) = 2x + 1, x \in \mathbb{R}$
- We focus on discrete objects: a discrete version
  - Think about the graph of $f(n) = 2n + 1, n \in \mathbb{Z}$
- In many case we could only draw a partial graph based on a subset S of the domain, denoted by $f(S)$
  - $\boldsymbol{f(S) = \{t | s \in S \wedge t = f(s)\}}$



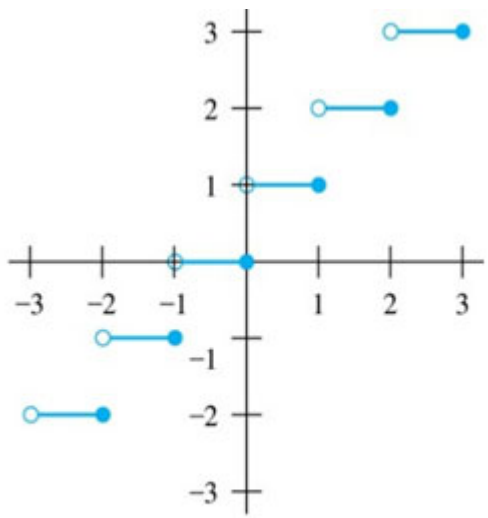
Graph of $f(x) = 2x + 1, x \in \mathbb{R}$ Graph of $f(n) = 2n + 1, n \in \mathbb{Z}$

Some Useful Functions

- You have seen functions to round up or round down a real number x: `floor()` and `ceil()`
  - `floor()` gives the largest integer that is less than or equal to x, can be denoted by $\lfloor x \rfloor$. E.g. $\lfloor 3.5 \rfloor = 3$, $\lfloor -3.5 \rfloor = -4$
  - `ceil()` gives smallest integer that is greater than or equal to x, can be denoted by $\lceil x \rceil$. E.g. $\lceil 3.5 \rceil = 4$, $\lceil -3.5 \rceil = -3$



(a) $y = \lfloor x \rfloor$



(b) $y = \lceil x \rceil$

- Optional properties

$$x - 1 < \lfloor x \rfloor \le x \le \lceil x \rceil < x + 1$$
$$-\lfloor x \rfloor = \lceil -x \rceil$$
$$-\lceil x \rceil = -\lfloor x \rfloor$$
$$\lfloor x + n \rfloor = \lfloor x \rfloor + n$$
$$\lceil x + n \rceil = \lceil x \rceil + n$$

Graph of floor function Graph of ceil function

Practice 1

- 1. Are these valid functions from $\mathbb{R}$ to $\mathbb{R}$?
  - $f(x) = \frac{1}{x}$
  - $f(x) = \sqrt{x}$
  - $f(x) = \pm\sqrt{x^2 + 1}$

Practice 2

- 1. Are these valid functions from $\mathbb{R}$ to $\mathbb{R}$?
  - $f(x) = \frac{1}{x}$
    - No. $f(0)$ is not defined: can't divide by 0.
  - $f(x) = \sqrt{x}$
    - No. $f(x)$ is not defined for negative real numbers.
  - $f(x) = \pm\sqrt{x^2 + 1}$
    - No. There are two output with different signs.

Practice 3

- 2. Determine whether each of these functions is invertible from $\mathbb{R}$ to $\mathbb{R}$.
  - $f(x) = 2x + 1$
  - $f(x) = x^2 + 1$
  - $f(x) = x^3$
  - $f(x) = (x^2 + 1)/(x^2 + 2)$

Practice 4

- 2. Determine whether each of these functions is invertible from $\mathbb{R}$ to $\mathbb{R}$.
  - $f(x) = 2x + 1$
    - Yes.
  - $f(x) = x^2 + 1$
    - No. Not one-to-one (1 and -1 mapped to 1), not onto (no output 0).
  - $f(x) = x^3$
    - Yes.
  - $f(x) = (x^2 + 1)/(x^2 + 2)$
    - No. Not one-to-one, not onto.

Practice 5

- 3. Find $(f \circ g)(x)$ and $(g \circ f)(x)$, where $f(x) = 2^x$ and $g(x) = 3x$ are functions from $\mathbb{R}$ to $\mathbb{R}$.

Practice 6

- 3. Find $(f \circ g)(x)$ and $(g \circ f)(x)$, where $f(x) = 2^x$ and $g(x) = 3x$ are functions from $\mathbb{R}$ to $\mathbb{R}$.
  - $(f \circ g)(x) = f\big(g(x)\big) = 2^{3x}$
  - $(g \circ f)(x) = g\big(f(x)\big) = 3 \cdot 2^x$
    - Please note: $2^{3x} = (2^x)^3$

Extension 1

- 1. Can you specify the properties of a function in propositional logic? For a function: $f: A \to B$
  - Full coverage of the domain / input

    $\forall a(a \in A \to \exists b(b \in B \wedge f(a) = b))$
  - No one-to-many mapping

    $\forall a, b_1, b_2(f(a) = b_1 \wedge f(a) = b_2 \to b_1 = b_2)$
  - No many-to-one mapping

    $\forall a_1, a_2\ (a_1 \neq a_2 \to f(a_1) \neq f(a_2))$
  - Only one-to-one mapping

    Combine "no one-to-many" and "no many-to-one"
  - Full coverage of the codomain

    $\forall b(b \in B \to \exists a(a \in A \wedge f(a) = b))$
- Inverse function of $f(n) = n^2: \mathbb{Z}^+ \to \mathbb{Z}^+$?

$f^{-1}(n) = \sqrt{n}: \mathbb{Z}^+ \to \mathbb{Z}^+$

Extension 2

- 2. Let A={a, b, c, d},B={1, 2, 3}. Define
  f(x): A→B where f(a)=2, f(b)=3, f(c)=2, f(d)=1
  g(x): B→A where g(1)=b, g(2)=d, g(3)=a.
- Are $(g \circ f)(x)$ and $(f \circ g)(x)$ invertible?
  - $k(x) = (g \circ f)(x) = g\big(f(x)\big)$: A → A with
    {(a, d), (b, a), (c, d), (d, b)}
    This is not onto, not one-to-one.
  - $h(x) = (f \circ g)(x) = f\big(g(x)\big)$: B → B with
    {(1, 3), (2, 1), (3, 2)}
    This is onto and one-to-one. Then $h^{-1}(x): B \to B$ with
    {(1, 2), (2, 3), (3, 1)}

Extension 3

- 3. Interview Practice A:
  - Corner case handling
  - Write a function similar to
    `std::string trim(const std::string& str)`
    that removes leading and trailing whitespace from a string.
    - Empty string: No crash, return empty
      - "" → ""
    - All whitespace: Distinguish leading/trailing vs all removal
      - " a " → “a"
    - Unicode whitespace: \t\n\r vs plain space
      - "\t\n a \r\n" → "a"
    - Embedded whitespace: Preserve middle spaces
      - " a b " → "a b"
    - Single character: Edge length
      - "\n" → ""

Extension 4

- 3. Interview Practice A:
  - Corner case handling
  - Write a function similar to
    `std::string trim(const std::string& str)`
    that removes leading and trailing whitespace from a string.

```
std::string trim(const std::string& str) {
    size_t start = str.find_first_not_of(" \t\n\r\f\v");
    if (start == std::string::npos) return "";  // All whitespace

    size_t end = str.find_last_not_of(" \t\n\r\f\v");
    return str.substr(start, end - start + 1);
}
// std::isspace() covers all of " \t\n\r\f\v“:  industry standard idiom.
```

Extension 5

- 4. Interview Practice B:
  - Corner case handling
  - Write a function:
    `int reverse(int x);`
    that reverses the digits of a 32-bit signed integer.
    - Negative numbers
      - -456 → -654
    - Trailing zeros
      - 120 → 21
    - Overflow
      - 1234567893 → 0 (overflow)
    - INT_MIN edge case
      - -2147434312 → -2134347412
    - Zero
      - 0 → 0
    - Single digit numbers
      - 1 → 1

Extension 6

- 4. Interview Practice B:
  - Corner case handling
  - Write a function:
    `int reverse(int x);`
    that reverses the digits of a 32-bit signed integer.

```
#include <climits>
int reverse(int x) {
    int result = 0;
    while (x != 0) {
        int digit = x % 10;
        x /= 10;

        // Check for overflow before multiplying
        if (result > INT_MAX / 10 ||
           (result == INT_MAX / 10 && digit > 7))
            return 0;
        if (result < INT_MIN / 10 ||
           (result == INT_MIN / 10 && digit < -8))
            return 0;
        result = result * 10 + digit;
    }
    return result;
}
```